

Signed arithmetic

Eight-bit two's complement treats the MSB as sign: range minus one twenty-eight through plus one twenty-seven

One hundred plus fifty starter

- Starter: add one hundred plus fifty in eight-bit two's complement
- Wrapped byte is hex nine six; signed view is minus one oh six
- Status panel shows true signed sum one fifty outside the eight-bit range
- Flags include V equals one when same-sign operands produce a result with flipped sign bit
- Compare with Example V equals one at one hundred plus one hundred
- And Example C equals one V equals zero at two hundred plus one hundred

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Signed arithmetic / overflow

8-bit two's complement add/sub — read **N Z C V** and see why carry $\neq$ overflow.

Starter example: 100 + 50 in 8-bit two's complement — compare a clean add with an overflowing one (e.g. 100+100).

Load starter example

Challenges (1/22)

Quiz: 2's. Signed encoding used here? Answer: two's complement

Answer

Show hintCheckNextIdle

quiz-twosquiz-vquiz-cquiz-rangestartercompute no-ov overflow carry-no-v neg-ov sub-ov op-sub

flag-zflag-ndemoexplaincode-vtrue-out hex-in wrap-neg c-vs-v full-path

Core ideas

Two's complement

MSB is sign; range -128...+127 for 8-bit.

Overflow V

Signed result doesn't fit — not the same as carry C.

Flags

N negative · Z zero · C carry/borrow · V overflow.

Signed arithmetic starter

learn_digital — signed arith

Workbook practice

- On paper, add plus one hundred plus plus fifty and list wrapped hex, signed view, and V
- Do one hundred plus one hundred and record minus fifty-six with V equals one
- Do two hundred plus one hundred and note C equals one with V equals zero
- Write the V rule for add: same sign in, different sign out
- Name one pitfall: treating C as signed overflow

Pitfalls to watch

- Do not read wrapped bits as the true signed answer when V equals one
- Subtract overflow uses a different V test than add, both mean out of range
- Negative numbers use two's complement, not sign-magnitude in this lab
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, run the starter and both overflow demos
- On paper, fill one row with operands, wrapped result, true signed, and NZCV
- When you are ready, take the short quiz, then continue to RAM and ROM map